

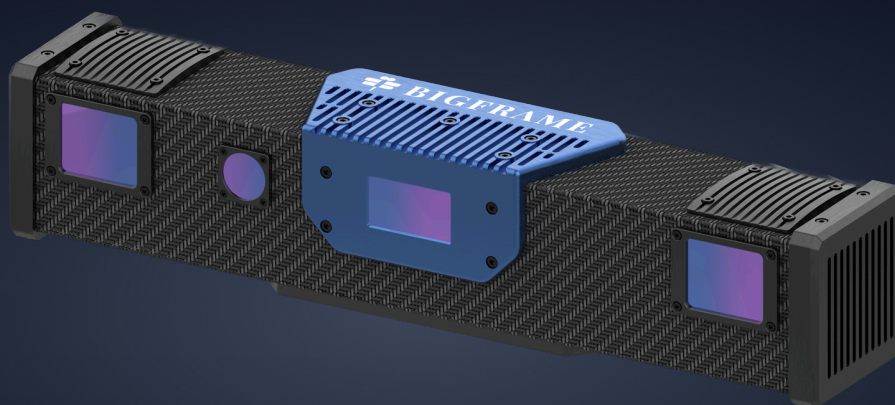
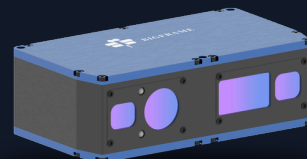
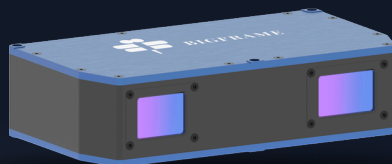
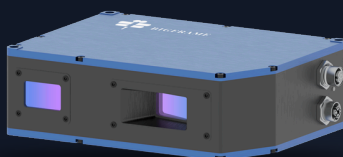
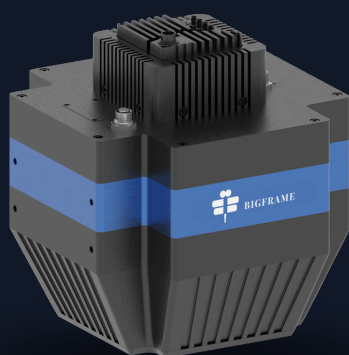


BIGFRAME

PRODUCT CATALOG 2026

Industrial 3D cameras

Eighteen structured-light 3D cameras in five series — fields of view from 9.8 mm to 3.6 m, for measurement, inspection and robot guidance.



TR S

Micrometer measurement

Solder balls, bumps, weld beads
FOV 9.8–74 mm

VR H 9

Fine measurement

Connectors, frames, boards
FOV 24–277 mm

VR

Guidance and inspection

Bins, pallets, body panels
FOV 0.3–1.5 m

VR D

Color stereo on the robot arm

Totes, tools, fasteners
FOV 0.58–1.17 m

PR S

NEW

Large scenes from a fixed bar

Pallets, vehicles, containers
FOV 1.3–3.6 m

The range at a glance

Eighteen structured-light 3D cameras in five series, with fields of view from 9.8 mm to 3.6 m. One software platform, TridiVision, across the range. Every figure is stated the same way: field of view at the near, optimum and far working distance, point spacing at the optimum, Z repeatability as the regional 2σ value (definitions and method, page 22).

CONTENTS			
01	Overview	2–4	The range · applications
02	Cameras	5–14	TRS · VRH9 · VR · VRD · PRS, two pages each
03	Software	15–16	TridiVision
04	Technical data	17–22	All models · series tables · accessories · definitions
05	Company	23	Bigframe · contact

TRS series

Micrometer measurement:
solder balls, bumps, weld beads

FOV 9.8–74 mm
Z repeatability from 0.06 μm

Page 5

VRH9 series

Fine measurement:
connectors, frames, boards

FOV 24–277 mm
Z repeatability from 0.15 μm

Page 7

VR series

Guidance and inspection:
bins, pallets, body panels

FOV 300–1500 mm
Z repeatability from 2 μm

Page 9

VRD series

Color stereo on the robot arm:
totes, tools, fasteners

FOV 580–1166 mm
Z repeatability from 6 μm

Page 11

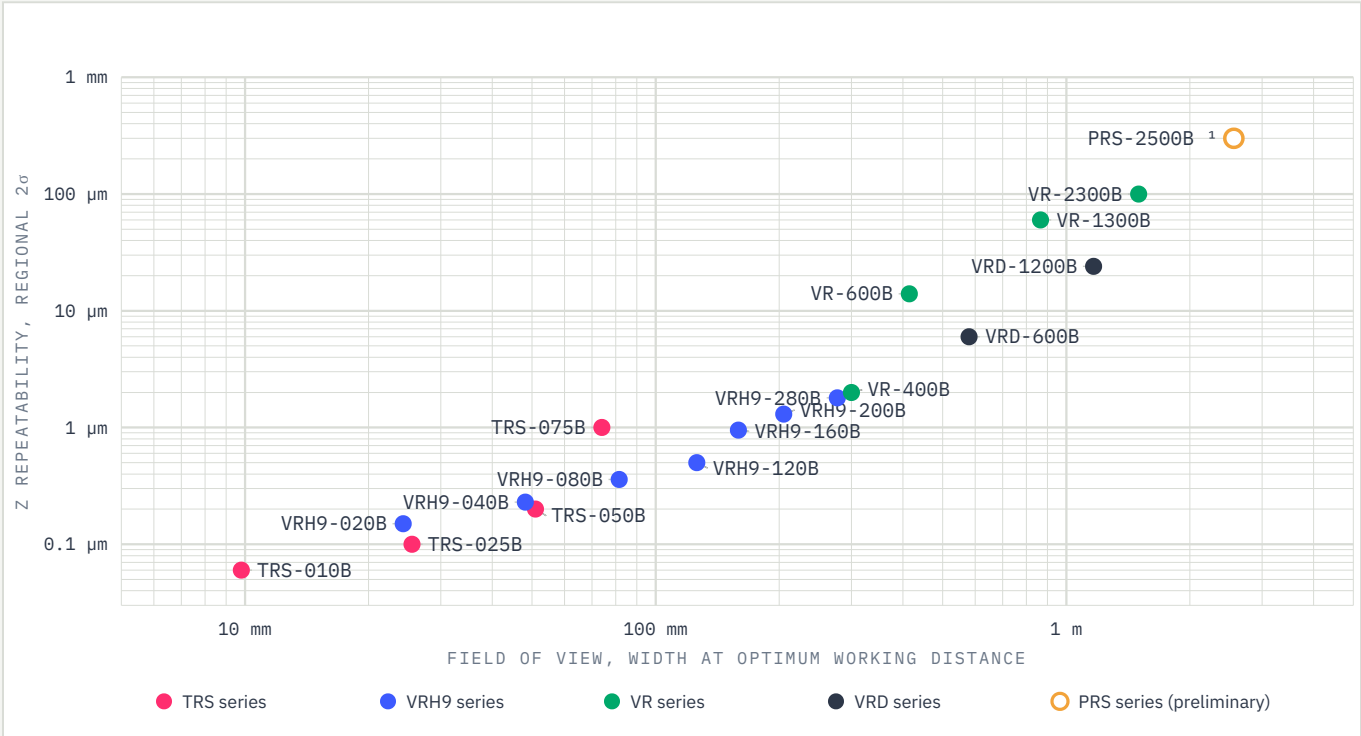
PRS series NEW

Large scenes from a fixed bar:
pallets, vehicles, containers

FOV 1.3–3.6 m at WD 1.2–3.5 m
Z repeatability 300 μm¹

Page 13

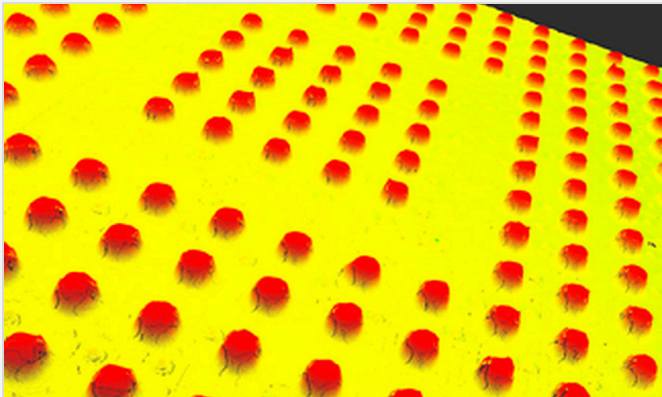
Field of view against Z repeatability



Width of the field of view at the optimum working distance against regional Z repeatability, 2σ, for all eighteen models. Adjacent series overlap: most parts have two candidate models. ¹ PRS: preliminary specification, page 13.

Inspection and measurement

Height, coplanarity, flatness and profile on the features that decide yield — from solder balls to body panels. Height maps from Bigframe application cases; the model codes list the cameras that cover each part.

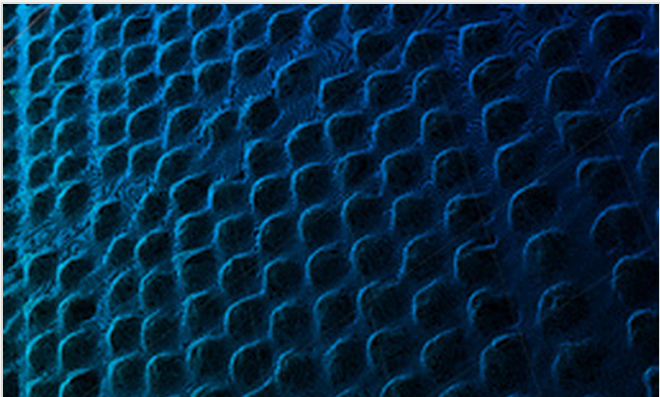


ELECTRONICS

BGA and CSP solder balls

Ball height, coplanarity, missing balls

TRS-010B · TRS-025B · VRH9-020B

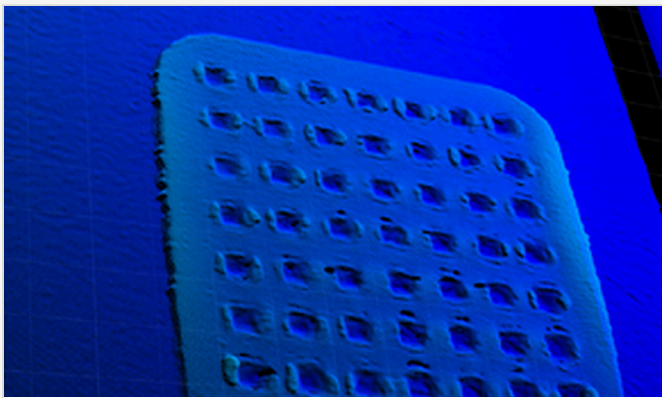


SEMICONDUCTOR

Wafer bumps

Bump height and coplanarity across the die

TRS-010B · TRS-025B

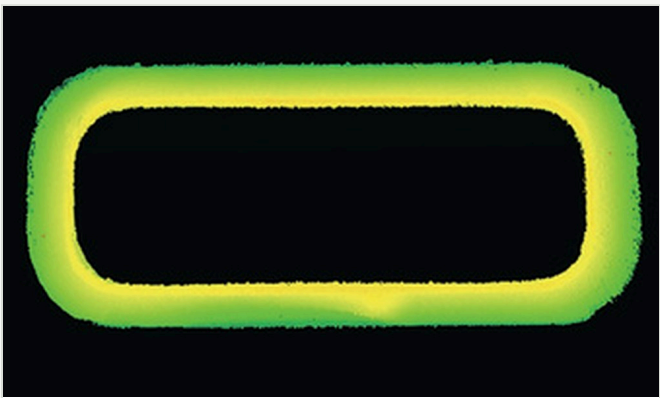


EV BATTERY

Cell contact system welds

Bead height and profile on every cell contact

TRS-050B · TRS-075B · VRH9-080B

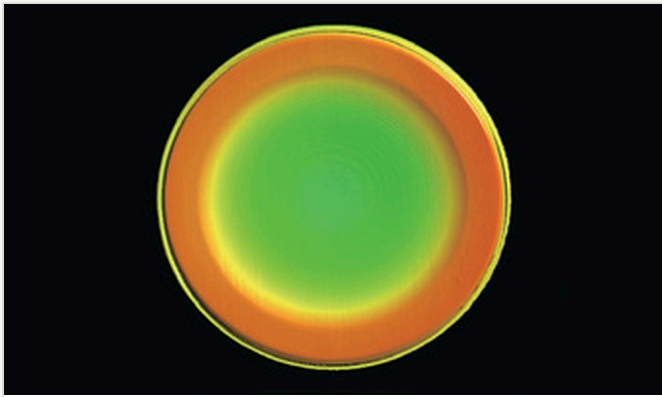


CONSUMER ELECTRONICS

Adhesive and sealing beads

Bead height and continuity on a USB-C connector

VRH9-020B · VRH9-040B

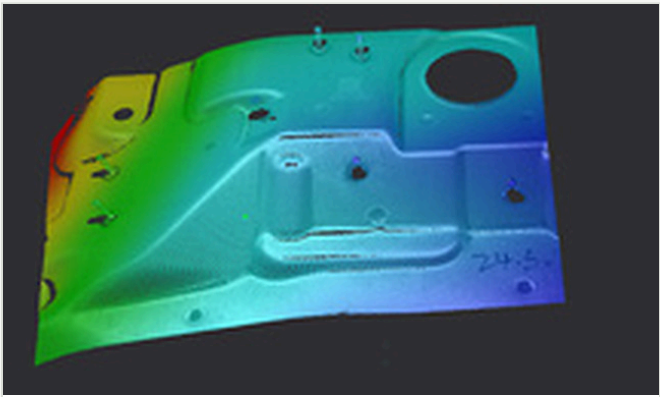


CONSUMER ELECTRONICS

Wireless charger housings

Surface profile and flatness of the ring

VRH9-080B · VRH9-120B



AUTOMOTIVE BODY

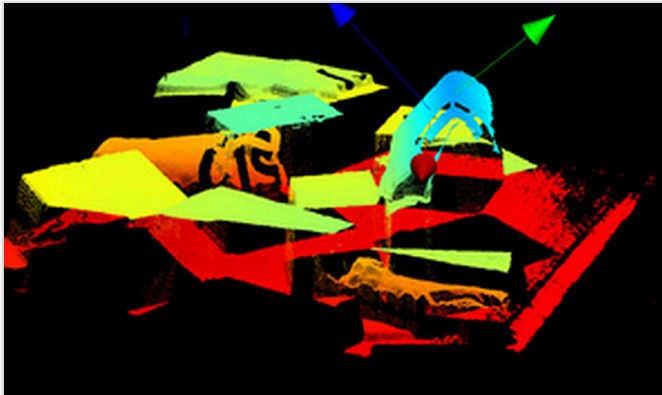
Stamped body panels

Hole position, surface and flange geometry

VR-600B · VR-1300B

Robot guidance and picking

Position and pose of parts in bins, totes and on pallets, from one stationary capture. Point clouds and color images from Bigframe application cases; the depalletizing scene is a rendering.

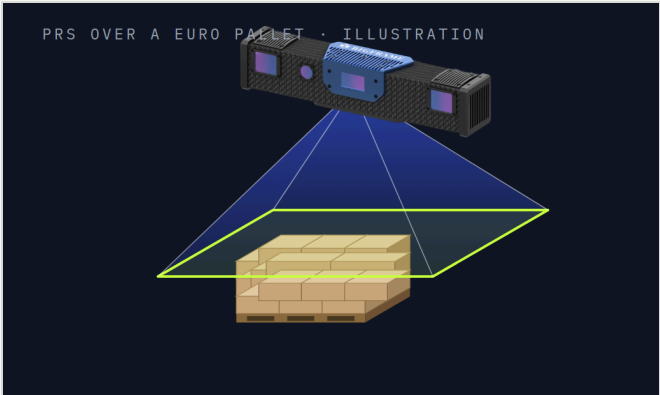


LOGISTICS

Parcel singulation and bin picking

Grasp points on unsorted parcels from the point cloud

VR-600B · VR-1300B · VRD-1200B



LOGISTICS

Depalletizing

Layer height and carton positions from one capture at 2.5 m

PRS-2500B · VR-2300B

Rendering — PRS over a Euro pallet



LOGISTICS · E-COMMERCE

Piece picking from totes

Items segmented by shape, color and print; pick pose per item

VRD-600B · VRD-1200B

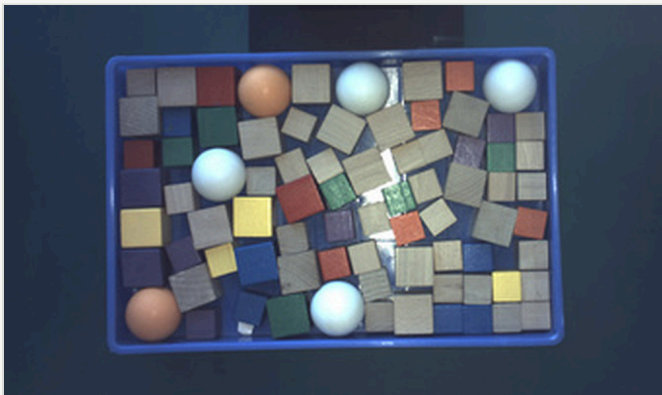


POWERTRAIN · MACHINE TENDING

Fastener bin picking

Screws, nuts and washers recognized in a mixed bin

VR-600B · VRD-600B



COLLABORATIVE ROBOTS

Sorting by color and shape

Parts segmented on the registered color image and picked

VRD-600B

YOUR APPLICATION

Not on these pages?

Send a photograph of the part, the feature to be measured and its tolerance. You receive the recommended model and working distance, and a clear statement of what the camera resolves — before you buy.

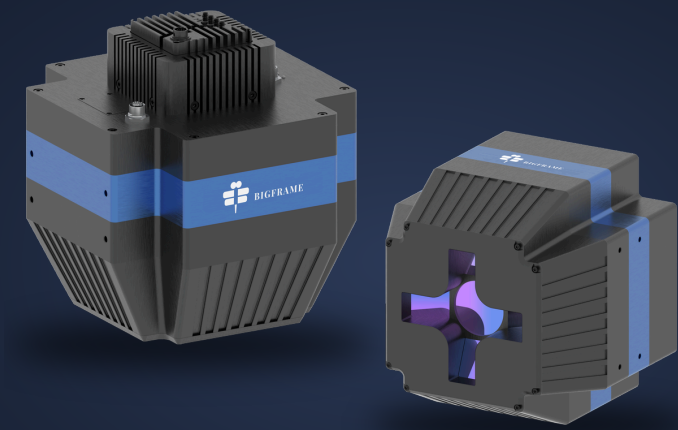
info@bigframe.com · www.bigframe.com



TRS SERIES

3D cameras for micrometer measurement

Solder balls, wafer bumps, terminals and weld beads; fields of view from 9.8 to 74 mm.



4

projection directions

Steps, holes and ball flanks lit from at least one side; fewer missing points.

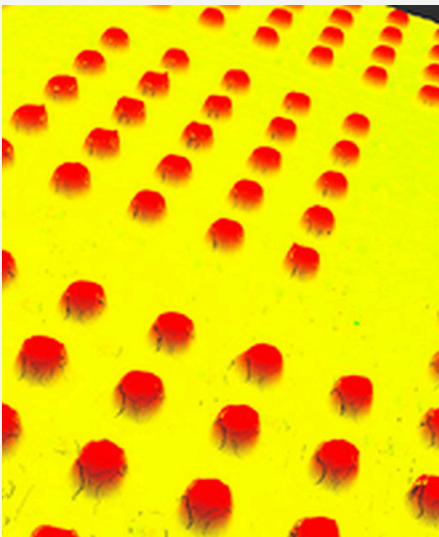
Four models, fields of view from 9.8 to 74 mm, point spacing from 1.9 to 14 μm : the scale at which yield is decided in semiconductor packaging and electronics assembly.

Models at a glance

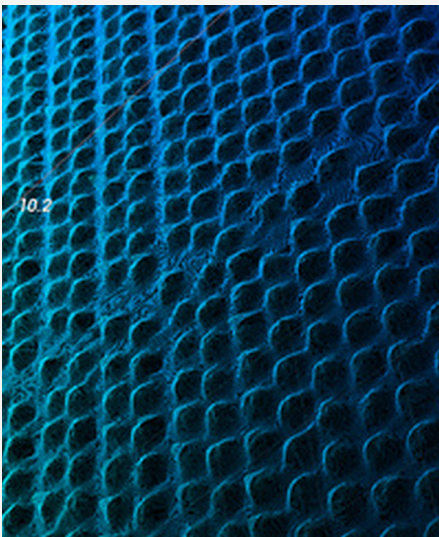
Model	Field of view at optimum WD (mm)	Working distance near / optimum / far (mm)	Measuring depth (mm)	XY point spacing (μm)	Z repeatability 2σ (μm) ¹	L × W × H (mm)	Weight (kg)
TRS-010B	9.8 × 9.8	62 / 62.5 / 63	1	1.9	0.06	230 × 230 × 250	8.85
TRS-025B	25.5 × 25.5	101.5 / 104 / 106.5	5	5	0.1	240 × 240 × 275	9.74
TRS-050B ²	51 × 51	– / 131 / –	8	10	0.2	300 × 300 × 253	11
TRS-075B ²	74 × 74	– / 191 / –	8	14	1	360 × 360 × 350	17

¹ Regional 2σ on a ceramic reference plate, method on page 22. Full specification on pages 18 to 20. ² Near and far working distances on request; measuring depth 8 mm.

Applications



BGA solder balls — ball height and coplanarity — height map

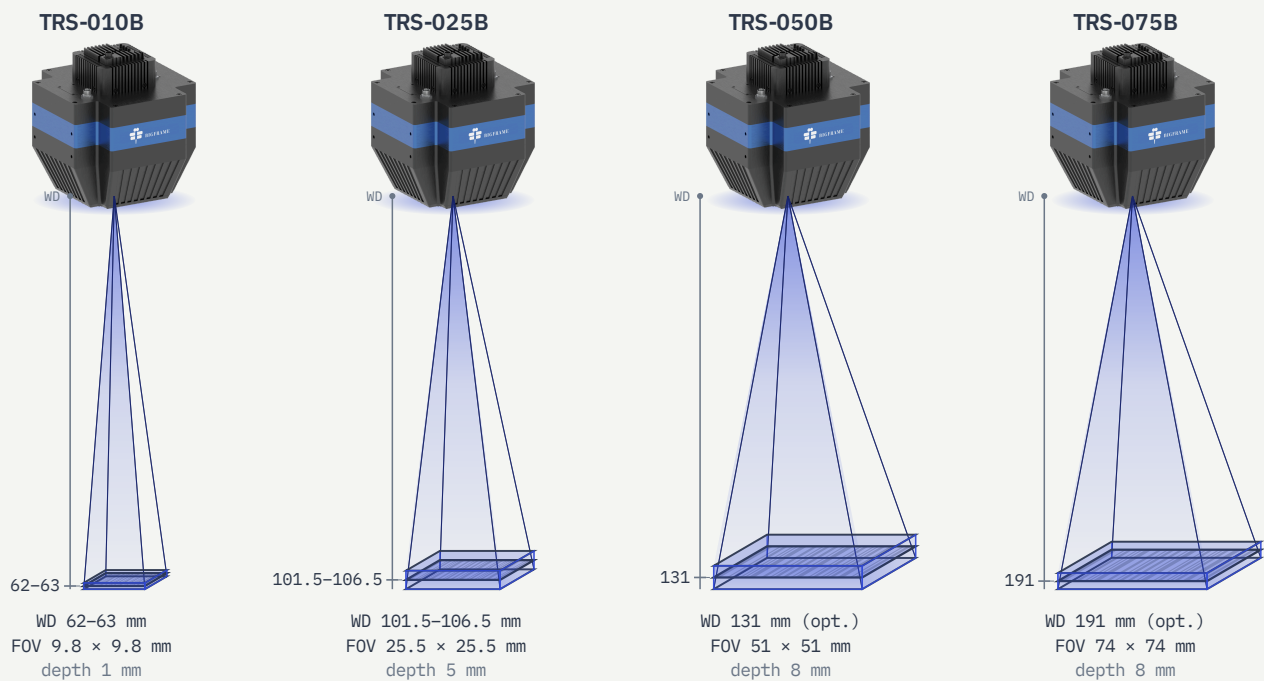


Wafer bumps — bump height and coplanarity across the die — height map

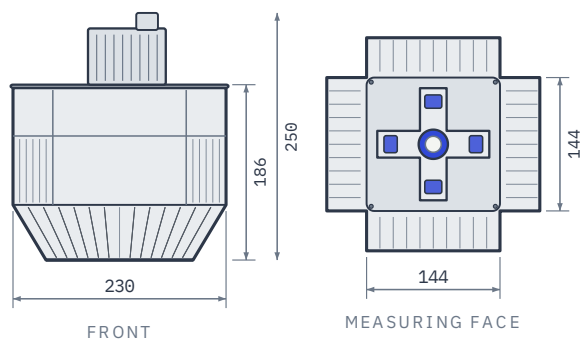


Cell contact system busbar welds — weld bead height and profile — height map

TRS series: field of view and dimensions



Cone angle and the proportions of near, optimum and far working distance are drawn to scale for each model; the models are not to scale with each other. Striped plane: field of view at the optimum working distance; translucent volume: measuring depth. Camera not to scale.

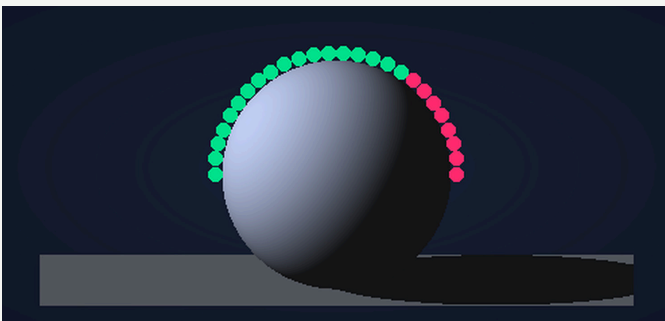


Dimensions and mounting

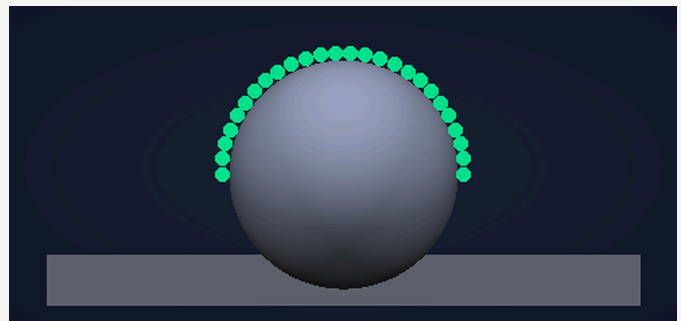
Housing L x W x H 230 x 230 x 250 mm (TRS-010B) to 360 x 360 x 350 mm (TRS-075B)
Weight 8.9 to 17 kg
Interface CoaXPress to a frame grabber; 24 V DC
Mounting 4 x M6 threads, 8 mm deep, on the side face

TRS-010B shown. The heat sink is part of the housing height; keep it free for convection. Working distances 62 to 191 mm. Dimension drawings and 3D CAD data on request.

Four projection directions: every flank measured



One projection direction: the far flank of the ball is in shadow and returns no points. Illustration.



Four projection directions: every flank is lit and measured. Illustration.

Your part, measured before you buy.

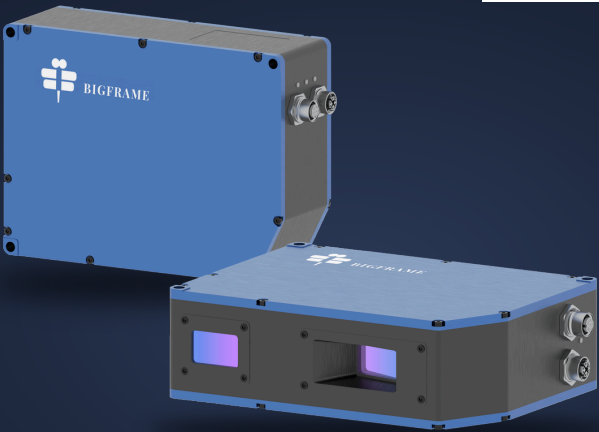
Send a photograph of the part, the feature and its tolerance to info@bigframe.com — we come back with the model and the working distance.



VRH9 SERIES

3D cameras for fine measurement, 24 to 277 mm fields of view

One housing, seven fields of view, for the parts that come off electronics and automotive lines.



24–277 mm

seven fields of view at optimum WD

One housing, one interface, one cable set and one integration for every model.

Connectors, mid-frames, adhesive beads, circuit assemblies, machined and stamped parts up to about 300 mm. One projector and one 4200 × 2160 camera in a 1.8 kg housing common to all seven models; the models differ only in optics.

9.07 million

points per frame

4200 × 2160 at 1.1 fps; 640 × 480 at 6.6 fps in guidance mode.

from 0.15 μm

Z repeatability, regional 2σ

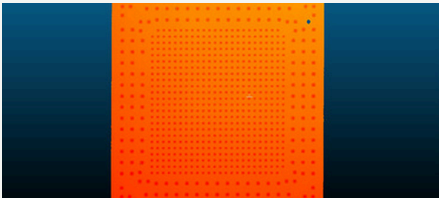
Measured as described on page 22. XY point spacing from 5.8 μm.

Models at a glance

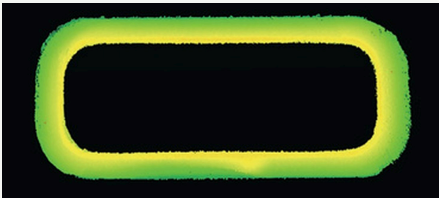
Model	Field of view at optimum WD (mm)	Working distance near / optimum / far (mm)	Measuring depth (mm)	XY point spacing (μm)	Z repeatability 2σ (μm) ¹
VRH9-020B	24.3 × 12.5	112 / 113 / 114.5	2.5	5.8	0.15
VRH9-040B	48.2 × 24.7	243 / 250 / 257	14	11	0.23
VRH9-080B	81.5 × 41.5	260 / 270 / 280	20	19	0.36
VRH9-120B	126 × 64.5	260 / 275 / 290	30	30	0.5
VRH9-160B	159 × 81.5	340 / 360 / 380	40	38	0.95
VRH9-200B	205 × 105	330 / 360 / 390	60	49	1.3
VRH9-280B	277 × 142	450 / 500 / 550	100	66	1.8

¹ Regional 2σ on a ceramic reference plate, method on page 22. Full specification on pages 18 to 20.

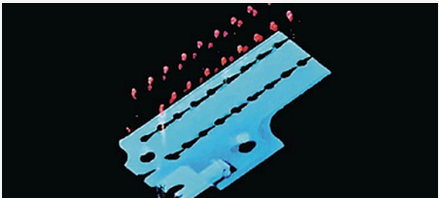
Applications



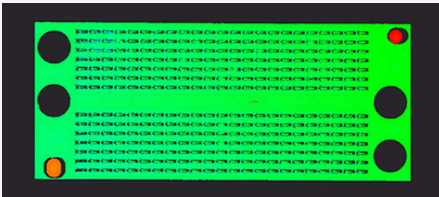
BGA solder balls — ball height and coplanarity — height map



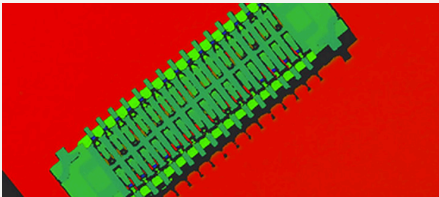
USB-C connector — sealing bead height and continuity — height map



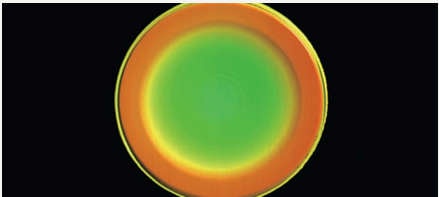
Connector pins — pin height and position — point cloud



Memory socket — spring-contact height and coplanarity — height map

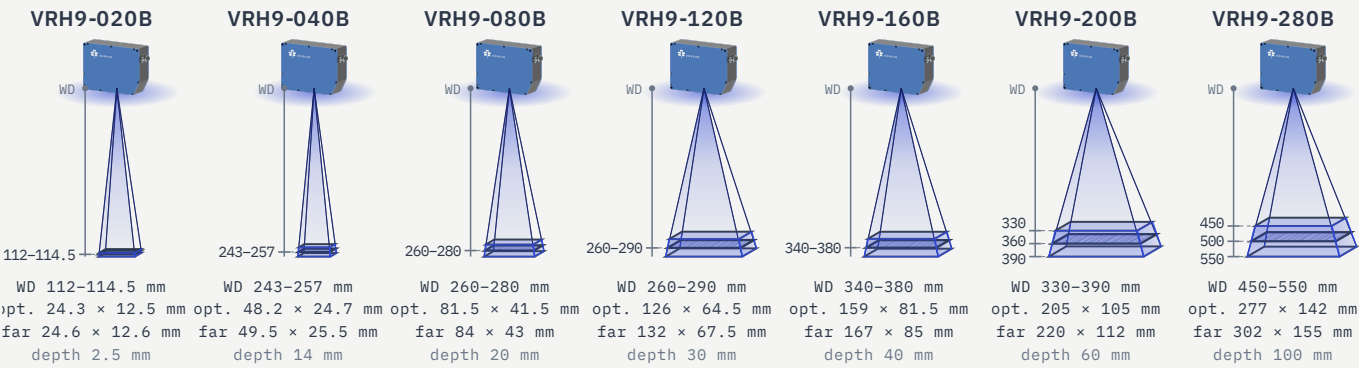


Board-to-board connector — terminal coplanarity and housing warp — height map

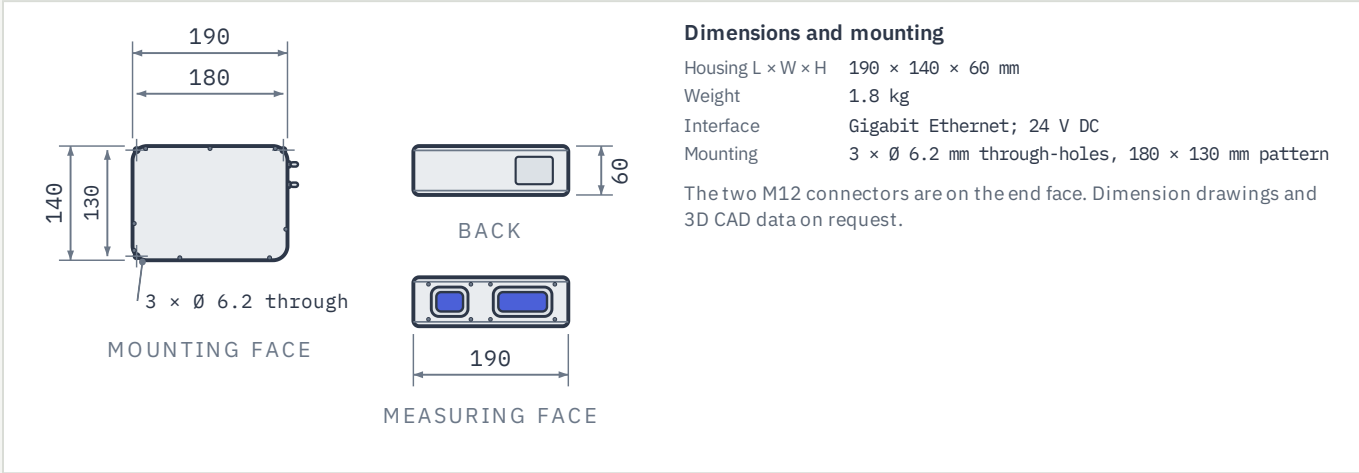


Wireless charger housing — surface profile and flatness of the ring — height map

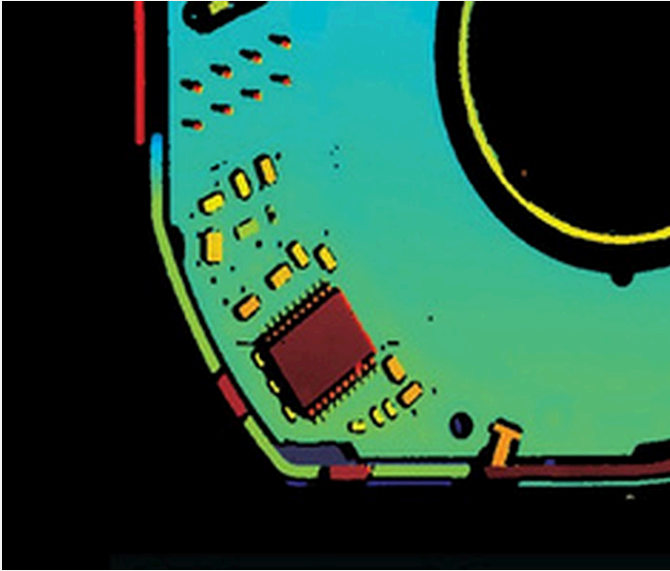
VRH9 series: field of view and dimensions



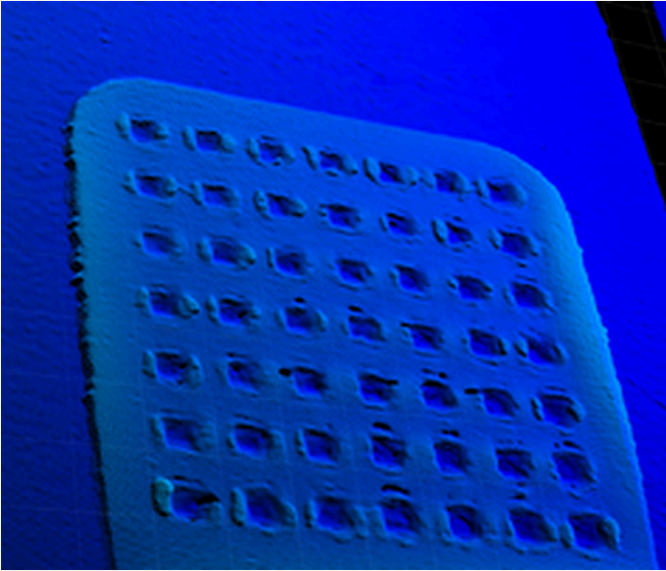
Cone angle and the proportions of near, optimum and far working distance are drawn to scale for each model; the models are not to scale with each other. Striped plane: field of view at the optimum working distance; translucent volume: measuring depth. Camera not to scale.



Applications



Printed circuit assembly — component presence and height, lifted leads — height map



Cell contact system welds — weld bead height and profile — height map

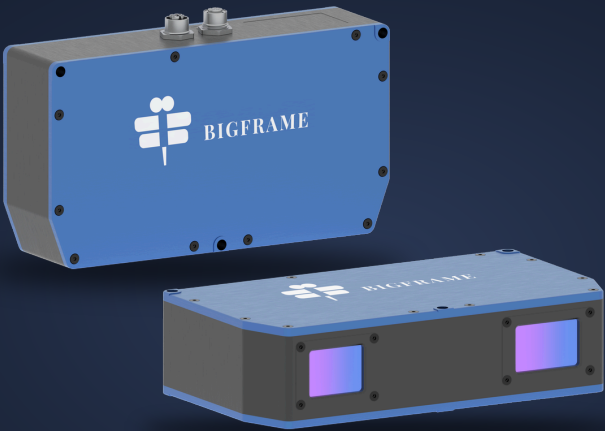
Your part, measured before you buy.

Send a photograph of the part, the feature and its tolerance to info@bigframe.com — we come back with the model and the working distance.

VR SERIES

3D cameras for bins, pallets and body panels

A bin, a pallet or a body panel from one stationary capture: guidance, picking, inspection.



300–1500 mm

field of view, four models

Working distances from 350 mm to 3.8 m; the same 1.6 kg housing for all four models.

One capture locates a parcel, a casting or a spindle and gives TridiVision the data to plan the robot path; the same cameras inspect sleepers, body panels and security trays.

3.2 million

points per frame

1.1 fps at full resolution; 4 fps at 640 × 480 in guidance mode.

from 2 μm

Z repeatability, regional 2σ

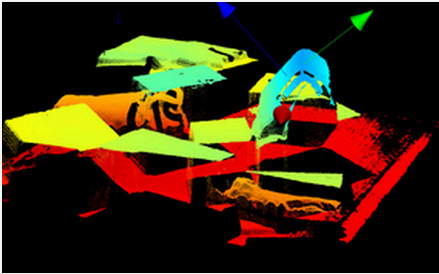
Measured as described on page 22. XY point spacing from 150 μm.

Models at a glance

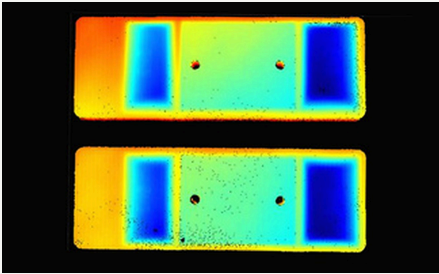
Model	Field of view at optimum WD (mm)	Working distance near / optimum / far (mm)	Measuring depth (mm)	XY point spacing (μm)	Z repeatability 2σ (μm) ¹
VR-400B	300 × 220	350 / 500 / 650	300	150	2
VR-600B	415 × 305	500 / 700 / 900	400	200	14
VR-1300B	865 × 645	900 / 1500 / 2200	1300	420	60
VR-2300B	1500 × 1115	1900 / 2600 / 3800	1900	730	100

¹ Regional 2σ on a ceramic reference plate, method on page 22. Full specification on pages 18 to 20.

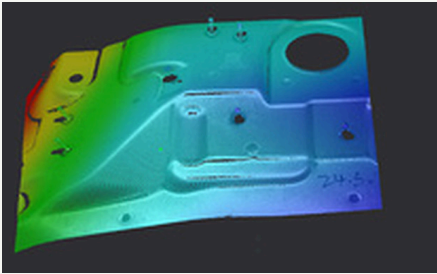
Applications



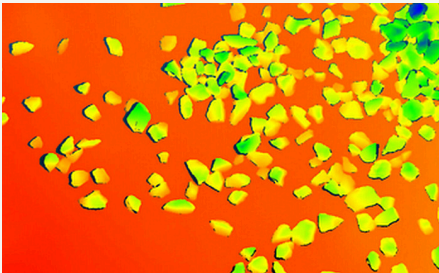
Mixed parcels in a bin — random bin picking — point cloud



Twin-block railway sleepers — block geometry — height map



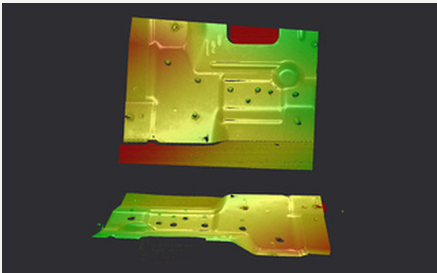
Stamped body panel — hole position and surface — height map



Crushed aggregate — particle size — height map

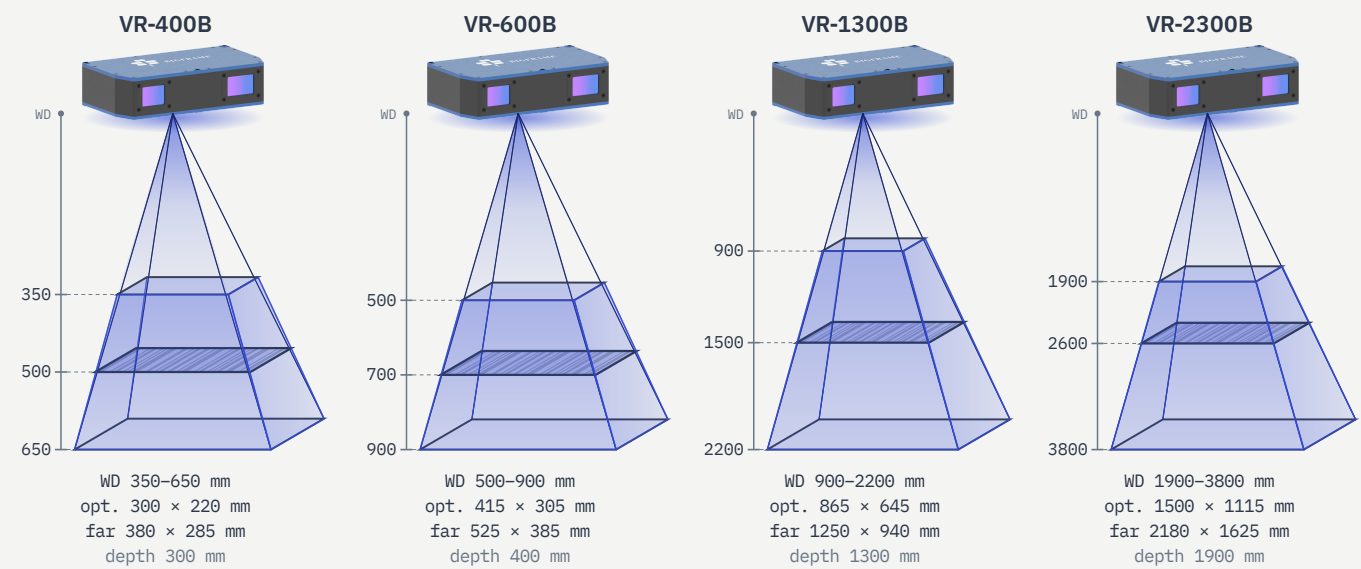


Yarn spindles in bulk — random picking — camera image

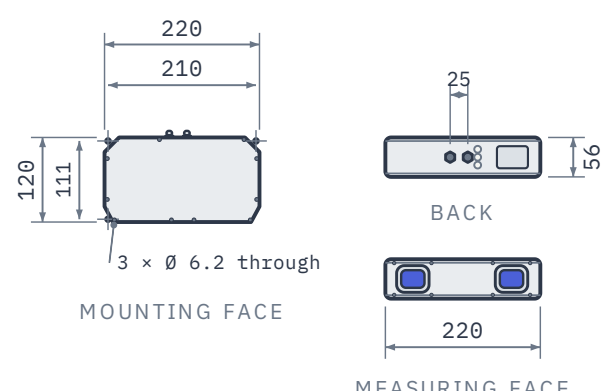


Stamped body panels — two panels in one capture — height map

VR series: field of view and dimensions



Cone angle and the proportions of near, optimum and far working distance are drawn to scale for each model; the models are not to scale with each other. Striped plane: field of view at the optimum working distance; translucent volume: measuring depth. Camera not to scale.



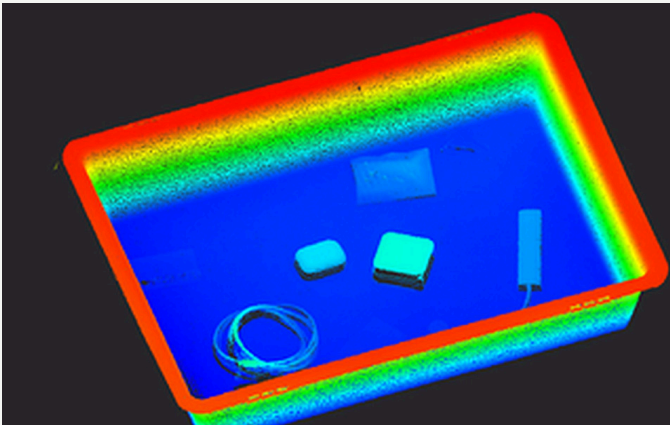
220
210
120
111
3 × Ø 6.2 through
MOUNTING FACE
BACK
MEASURING FACE

Dimensions and mounting

Housing L × W × H	220 × 120 × 56 mm
Weight	1.6 kg
Interface	Gigabit Ethernet; 24 V DC
Mounting	3 × Ø 6.2 mm through-holes, 210 × 111 mm pattern

The two M12 connectors are on the back face. Dimension drawings and 3D CAD data on request.

Applications



Airport security trays — left-behind items — height map



Fuel filler neck — nozzle insertion guided from the height map

Your bin, pallet or panel,
measured before you buy.

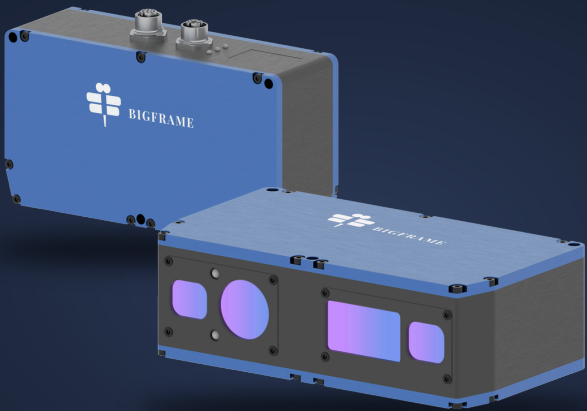
Send a photograph of the scene and the measurement you need to info@bigframe.com — we come back with the model and the working distance.



VRD SERIES

Color stereo 3D cameras for collaborative robots

An 870 g eye-in-hand camera: point cloud and registered color image in one capture.



870 g

eye-in-hand weight

Mounted on the flange of collaborative robots; two models, 600 mm and 1200 mm optimum working distance.

The registered color image lets TridiVision segment mixed parcels, packaged goods, tools and fasteners by color and printed markings as well as by shape; the point cloud gives the pick pose.

5 million

points per frame, with RGB

A point cloud and a registered color image from every capture.

from 6 μm

Z repeatability, regional 2σ

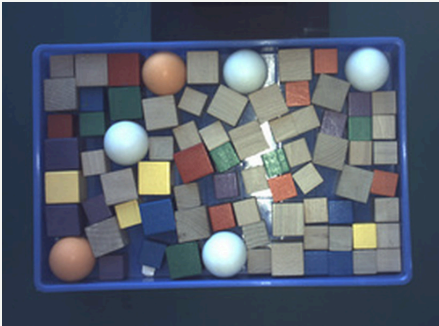
Measured as described on page 22. XY point spacing from 240 μm.

Models at a glance

Model	Field of view at optimum WD (mm)	Working distance near / optimum / far (mm)	Measuring depth (mm)	XY point spacing (μm)	Z repeatability 2σ (μm) ¹
VRD-600B	580 × 369	400 / 600 / 800	400	240	6
VRD-1200B	1166 × 731	900 / 1200 / 1600	700	480	24

¹ Regional 2σ on a ceramic reference plate, method on page 22. Full specification on pages 18 to 20.

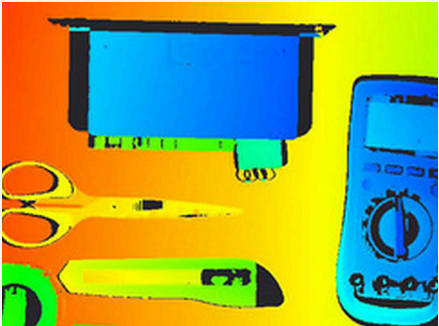
Applications



Colored blocks and balls in a tote — sorting by color and shape — color image



Hand tools in a bin — recognition and picking — color image



Hand tools — the same capture — height map



Mixed items in a tote — piece picking — color image

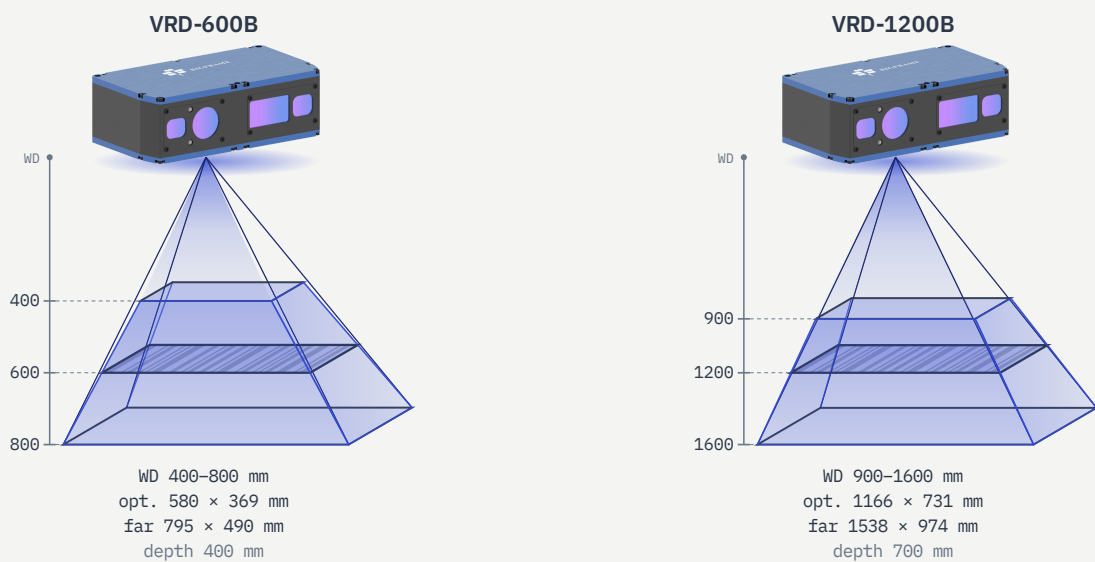


Screws, nuts and washers — mixed fastener picking — color image



Packaged goods — picking by print and shape — color image

VRD series: field of view and dimensions



Cone angle and the proportions of near, optimum and far working distance are drawn to scale for each model; the models are not to scale with each other. Striped plane: field of view at the optimum working distance; translucent volume: measuring depth. Camera not to scale.

166
156
90
82
3 × Ø 5.2 through
MOUNTING FACE
BACK
MEASURING FACE

Dimensions and mounting

Housing L × W × H

166 × 90 × 54 mm

Weight

870 g

Interface

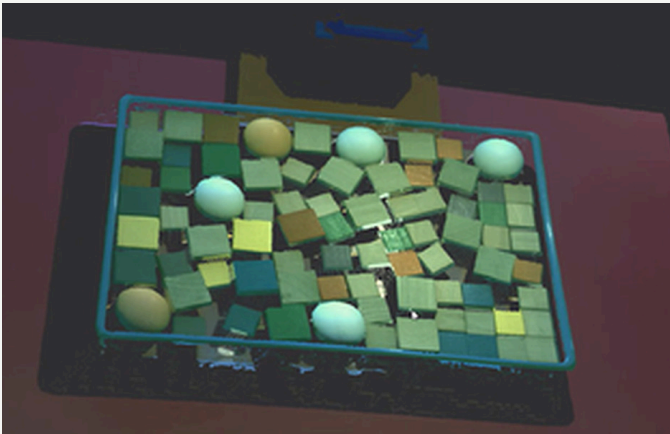
Gigabit Ethernet; 24 V DC

Mounting

3 × Ø 5.2 mm through-holes, 156 × 82 mm pattern

The two M12 connectors are on the back face. Dimension drawings and 3D CAD data on request.

Applications



A tote of colored blocks from the working distance of the camera — color image



Mixed items in a tote, second scene — piece picking — color image

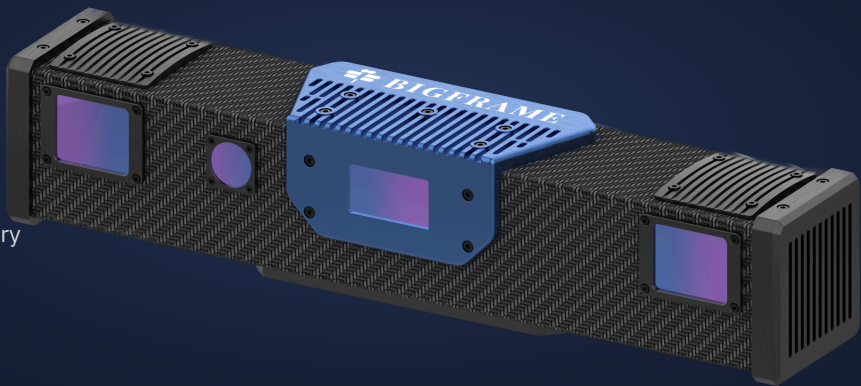
Your picking cell, checked before you buy.

Send a photograph of the parts and the tote to info@bigframe.com — we come back with the model and the working distance.

3D cameras for pallets, vehicles and containers

Large scenes at 1.2 to 3.5 m from a bar above the dock, the cell or the conveyor; a color image registered to every point.

PRELIMINARY · VISION 2026



2.56 × 2.06 m

field of view at 2.5 m

1.30 × 1.05 m at 1.2 m; 3.60 × 2.91 m at 3.5 m.

1.2–3.5 m

working distance

2.3 m measuring depth from one fixed mount; 890 mm bar, 3.9 kg, above the scene.

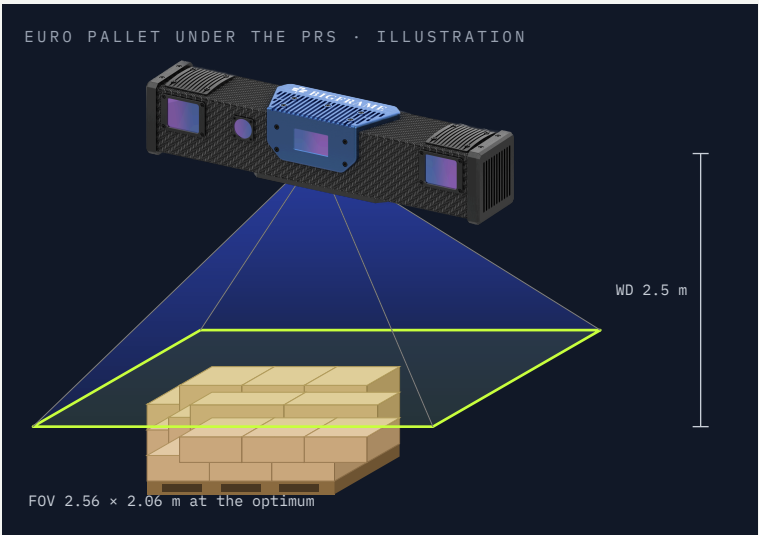
300 μm

Z repeatability ¹

With 1400 μm point spacing at 2.5 m; two 5 MP cameras, RGB image, 1 fps.

The PRS covers the largest scenes in the range — a pallet, a vehicle, a container, a stack on a dock — from one capture, with the color image the operator needs to see what was measured.

The PRS in use



At 2.5 m the field of view of 2.56 × 2.06 m covers a Euro pallet with margin; layer height and carton positions come from one capture.

Which large-scene camera?

Choose the VR-2300B where the finest point spacing at 2 to 4 m matters and no color image is needed; the VRD-1200B for a camera on the arm; the PRS where a scene of 2 m or more must be covered from a fixed mount with a color image.

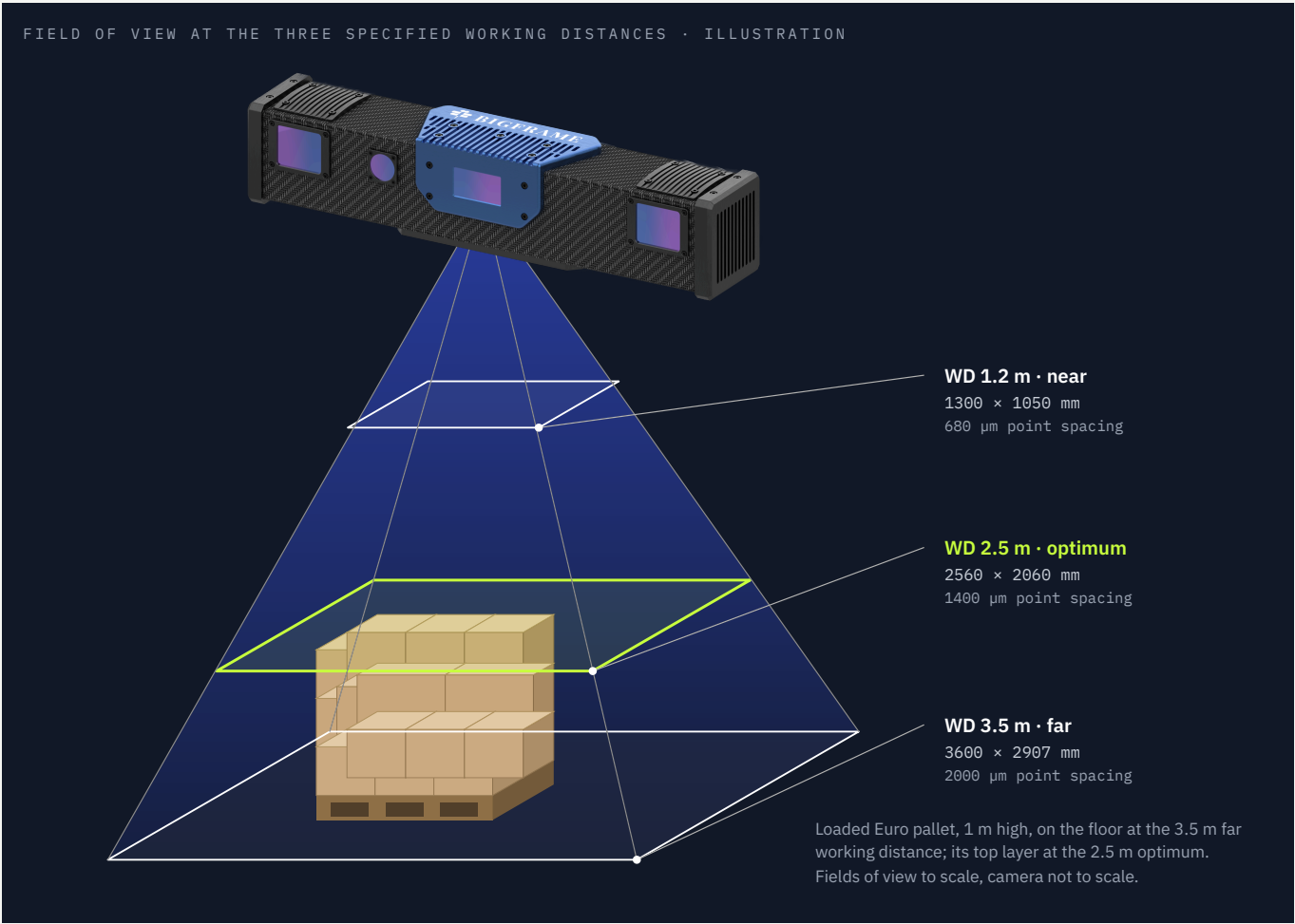
The PRS-2500B is specified at three working distances from 1.2 to 3.5 m (table on page 14); at 3.5 m the field of view of 3.60 × 2.91 m covers a container door with margin; at 1.2 m it is 1.30 × 1.05 m.

Large-scene cameras compared

Model	Field of view at optimum WD (mm)	Working distance near / optimum / far (mm)	XY point spacing (μm)	Z repeatability 2σ (μm)	Color image	Mounting
VR-1300B	865 × 645	900 / 1500 / 2200	420	60	No	fixed, above the scene
VR-2300B	1500 × 1115	1900 / 2600 / 3800	730	100	No	fixed, above the scene
VRD-1200B	1166 × 731	900 / 1200 / 1600	480	24	Yes	robot arm
PRS-2500B ¹	2560 × 2060	1200 / 2500 / 3500	1400	300	Yes	fixed, above the scene

¹ PRS: preliminary specification; Z repeatability 300 μm, method to be stated at launch.

PRS series: field of view and dimensions

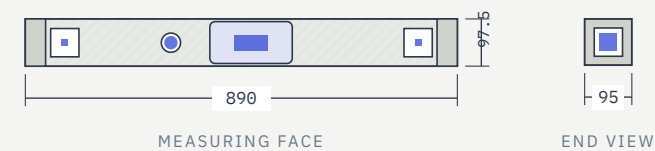


Field of view by working distance ¹

WD (mm)	Field of view (mm)	XY point spacing (µm)
1200 near	1300 × 1050	680
2500 opt.	2560 × 2060	1400
3500 far	3600 × 2907	2000

¹ Preliminary specification; final data sheet at launch. Z repeatability 300 µm, method to be stated at launch. Measuring depth 2300 mm, near to far.

Dimensions and mounting



Housing L × W × H	890 × 97.5 × 95 mm
Weight	3.9 kg
Stereo baseline	approx. 300 mm
Interface	Gigabit Ethernet; 24 V DC, up to 4 A
Mounting	Bracket kit; details at launch

Preliminary specification; dimensions over the housing, connectors excluded. Dimension drawings and 3D CAD data on request.

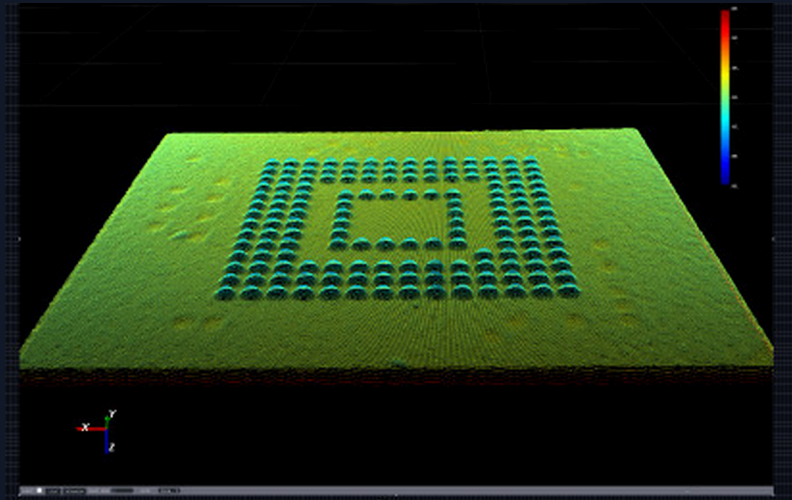
Your pallet or vehicle,
measured before you buy.

Send a photograph of the scene and the measurement you need to info@bigframe.com — we come back with the model and the working distance.

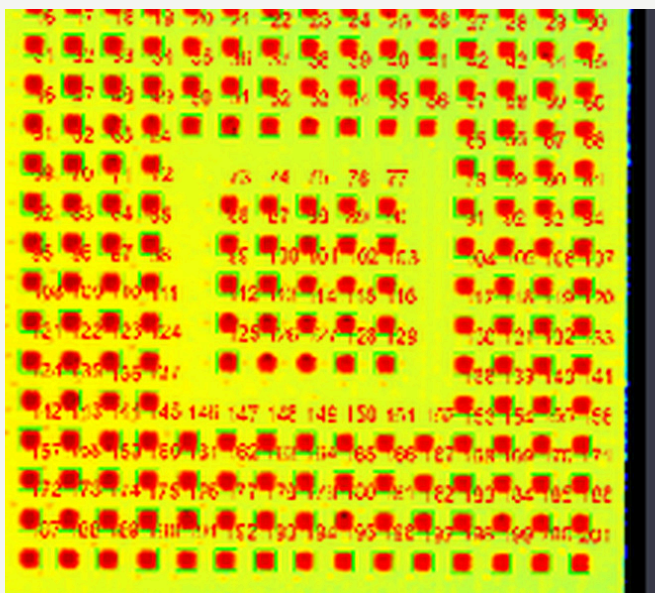
SOFTWARE

TridiVision

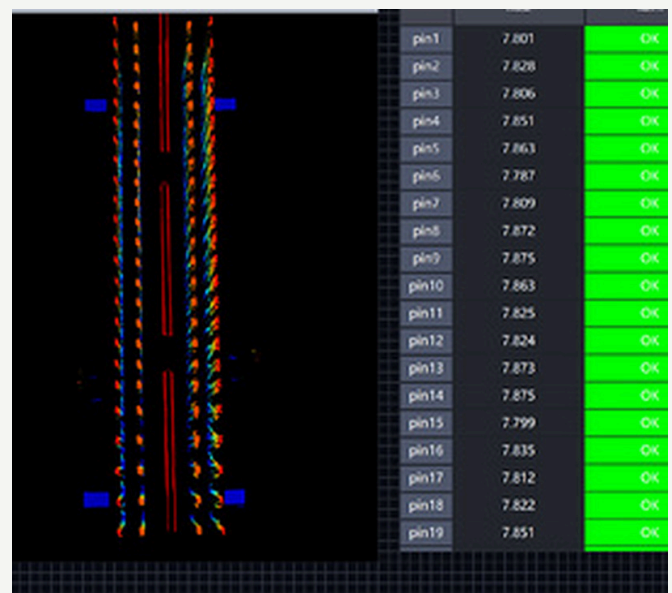
The vision software for every Bigframe camera: 2D, 3D and deep learning in one graphical environment.



A graphical dataflow environment: place tools on a canvas, link their parameters, run them and read the results. Standard tasks need no code; custom operators can be added for the rest. The same environment handles 2D images, height maps and point clouds, runs deep-learning models trained in its AI Training edition, builds the operator screen and drives third-party 2D and 3D cameras alongside the Bigframe camera. Results, signals and robot paths go to the PLC, MES or robot.



Solder-ball inspection — every ball numbered and measured on the canvas



Pin inspection — height of every pin and its pass/fail state in the operator screen

Function overview

2D

Positioning, measurement, defect detection, OCR and code reading, image processing, calibration.

Deep learning

Classification, segmentation, detection; AI Runtime runs the models, AI Training trains them.

3D Z-map

Height, step, flatness and coplanarity, distance and angle on height maps; multi-camera stitching.

Communication

PLC, MES and robot interfaces; results, signals and robot paths.

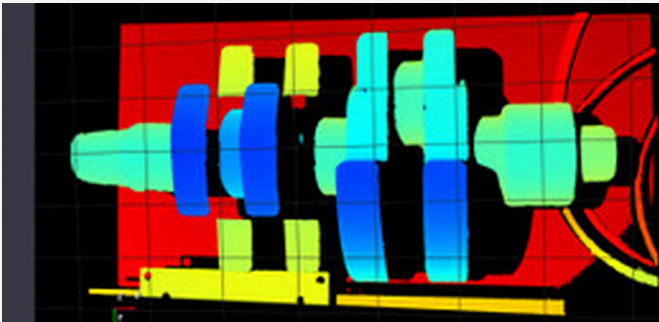
3D point cloud

Recognition and pose, robot path planning, point-cloud processing and export.

Operator screen

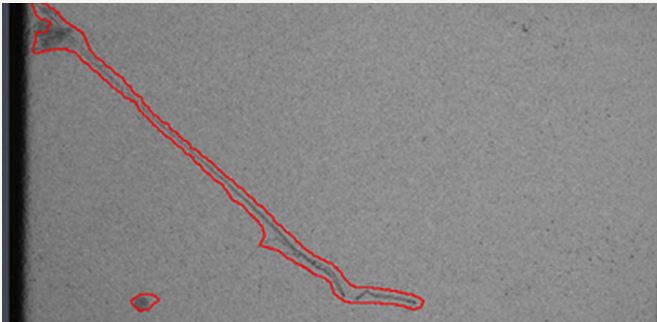
HMI designer: build the operator screen by drag and drop.

TridiVision: functions and editions



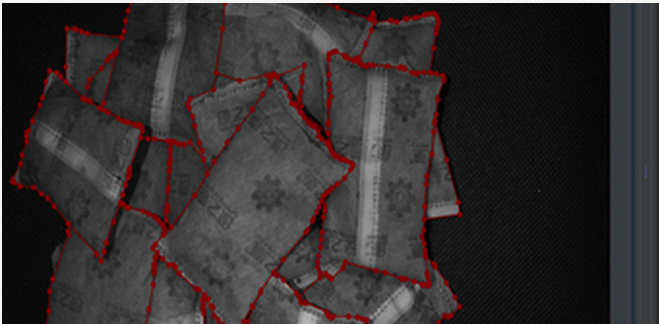
3D measurement tools

Height, step, flatness and coplanarity, distance and angle on point clouds and height maps; features fitted on the cloud; multi-camera stitching.



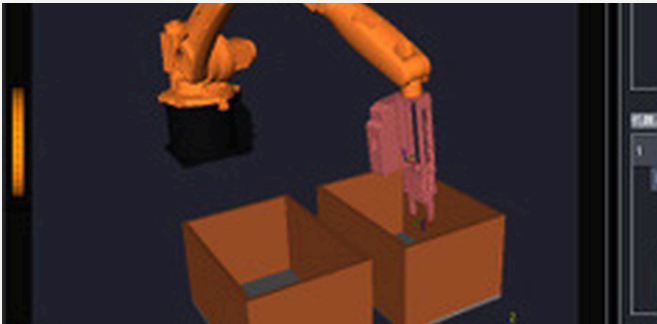
Deep learning

Classification, semantic segmentation and object detection — models trained in AI Training, run in AI Runtime.



Annotation and training

Annotation and training from a small image set; camera, hand-eye and multi-camera calibration on the Bigframe targets.



Robots and communication

Object recognition with grasp guidance and path planning from the point cloud; results and signals to PLCs, MES and robots.

Editions

Five editions on a USB hardware key; each includes everything in the edition before it.

Edition	USB key	Modules included
TridiVision 2D	● red	1D inspection; 2D positioning and measurement; image processing; defect detection; calibration; computation; communication
TridiVision 3D Z-map	● orange	everything in 2D, plus 3D Z-map positioning, measurement and processing
TridiVision 3D Point Cloud	● yellow	everything in 3D Z-map, plus 3D recognition, robot path planning and point-cloud processing
TridiVision AI Runtime	● green	everything in 3D Point Cloud, plus deep-learning inference
TridiVision AI Training (all-in-one)	● cyan	everything in AI Runtime, plus deep-learning training

System requirements

GPU acceleration with NVIDIA CUDA; a CoaXPress frame grabber in the host computer for the TRS series. Operating system, memory and GPU as stated in the installation guide. Sample programs for typical applications ship with the software.

See TridiVision on your parts.

Send sample images or a part to info@bigframe.com: we build the inspection in TridiVision and show it running on your data.

All models

The figures that decide the choice, in catalog order. Full specifications on pages 18 to 20; definitions and the repeatability method on page 22.

Model	Field of view at optimum WD (mm)	Working distance near–far (mm)	Optimum WD (mm)	Measuring depth (mm)	XY point spacing (μm)	Z repeatability 2σ (μm) ¹	Points per frame	Interface	L × W × H (mm)	Weight (kg)	Page
● TRS series — Four-projection structured-light 3D cameras											
TRS-010B	9.8 × 9.8	62–63	62.5	1	1.9	0.06	26.2 million	CoaXPress	230 × 230 × 250	8.85	5
TRS-025B	25.5 × 25.5	101.5–106.5	104	5	5	0.1	26.2 million	CoaXPress	240 × 240 × 275	9.74	5
TRS-050B ²	51 × 51	on request	131	8	10	0.2	26.2 million	CoaXPress	300 × 300 × 253	11	5
TRS-075B ²	74 × 74	on request	191	8	14	1	26.2 million	CoaXPress	360 × 360 × 350	17	5
● VRH9 series — Area-scan structured-light 3D cameras, fine measurement											
VRH9-020B	24.3 × 12.5	112–114.5	113	2.5	5.8	0.15	9.07 million	GigE	190 × 140 × 60	1.8	7
VRH9-040B	48.2 × 24.7	243–257	250	14	11	0.23	9.07 million	GigE	190 × 140 × 60	1.8	7
VRH9-080B	81.5 × 41.5	260–280	270	20	19	0.36	9.07 million	GigE	190 × 140 × 60	1.8	7
VRH9-120B	126 × 64.5	260–290	275	30	30	0.5	9.07 million	GigE	190 × 140 × 60	1.8	7
VRH9-160B	159 × 81.5	340–380	360	40	38	0.95	9.07 million	GigE	190 × 140 × 60	1.8	7
VRH9-200B	205 × 105	330–390	360	60	49	1.3	9.07 million	GigE	190 × 140 × 60	1.8	7
VRH9-280B	277 × 142	450–550	500	100	66	1.8	9.07 million	GigE	190 × 140 × 60	1.8	7
● VR series — Area-scan structured-light 3D cameras, medium and large scenes											
VR-400B	300 × 220	350–650	500	300	150	2	3.2 million	GigE	220 × 120 × 56	1.6	9
VR-600B	415 × 305	500–900	700	400	200	14	3.2 million	GigE	220 × 120 × 56	1.6	9
VR-1300B	865 × 645	900–2200	1500	1300	420	60	3.2 million	GigE	220 × 120 × 56	1.6	9
VR-2300B	1500 × 1115	1900–3800	2600	1900	730	100	3.2 million	GigE	220 × 120 × 56	1.6	9
● VRD series — Color stereo structured-light 3D cameras, robot guidance											
VRD-600B	580 × 369	400–800	600	400	240	6	5 million	GigE	166 × 90 × 54	0.87	11
VRD-1200B	1166 × 731	900–1600	1200	700	480	24	5 million	GigE	166 × 90 × 54	0.87	11
● PRS series — Wide-baseline stereo 3D cameras for large scenes · preliminary ³											
PRS-2500B ³	2560 × 2060	1200–3500	2500	2300	1400	300	on request	GigE	890 × 97.5 × 95	3.9	13

¹ Regional 2σ on a ceramic reference plate; the largest value found at the near, optimum and far working distance; method on page 22. ² TRS-050B and TRS-075B: near and far working distances on request; measuring depth 8 mm. ³ PRS: preliminary specification; Z repeatability method and points per frame stated at launch.

Selecting a model

Choose a **field of view** that covers the part with margin at the optimum working distance.

Choose a **point spacing** of one tenth of the smallest feature or finer.

Choose a **Z repeatability** of one tenth of the tolerance or better; a gauge study on the part decides.

Which model? Ask us.

Send a photograph of the part, the feature and its tolerance to info@bigframe.com. We come back with the model and the working distance, and say what the camera will and will not resolve.



TRS and VRD series: technical data

TRS series

Model	TRS-010B	TRS-025B	TRS-050B ²	TRS-075B ²
Working distance, near (mm)	62	101.5	–	–
Working distance, optimum (mm)	62.5	104	131	191
Working distance, far (mm)	63	106.5	–	–
Field of view at near WD, W × H (mm)	9.8 × 9.8	25.5 × 25.5	–	–
Field of view at optimum WD, W × H (mm)	9.8 × 9.8	25.5 × 25.5	51 × 51	74 × 74
Field of view at far WD, W × H (mm)	9.8 × 9.8	25.5 × 25.5	–	–
Measuring depth, Z range (mm)	1	5	8	8
Optimum measuring volume, W × H × D (mm)	9.8 × 9.8 × 1	25.5 × 25.5 × 5	51 × 51 × 8	74 × 74 × 8
XY point spacing at optimum WD (μm)	1.9	5	10	14
Z repeatability, regional 2σ (μm) ¹	0.06	0.1	0.2	1
Dimensions L × W × H (mm)	230 × 230 × 250	240 × 240 × 275	300 × 300 × 253	360 × 360 × 350
Weight (kg)	8.85	9.74	11	17

COMMON TO ALL MODELS OF THE SERIES			
3D points per frame	26.2 million (5120 × 5120)	Frame rate	Up to 3 fps at 5120 × 5120
Optical configuration	Four projection directions, one camera	Light source	Blue light
Color image output	No	Data interface	CoaXPress
Power	24 V DC; power supply rated 75 W or more	Software	TridiVision

¹ Regional Z repeatability, 2σ, measured by the method on page 22; the largest value found at the near, optimum and far working distance. ² TRS-050B and TRS-075B: near and far working distances and fields of view on request; measuring depth 8 mm.

VRD series

Model	VRD-600B	VRD-1200B
Working distance, near (mm)	400	900
Working distance, optimum (mm)	600	1200
Working distance, far (mm)	800	1600
Field of view at near WD, W × H (mm)	365 × 248	887 × 551
Field of view at optimum WD, W × H (mm)	580 × 369	1166 × 731
Field of view at far WD, W × H (mm)	795 × 490	1538 × 974
Measuring depth, Z range (mm)	400	700
Optimum measuring volume, W × H × D (mm) ³	580 × 369 × 300	1166 × 731 × 400
XY point spacing at optimum WD (μm)	240	480
Z repeatability, regional 2σ (μm) ¹	6	24

COMMON TO ALL MODELS OF THE SERIES			
3D points per frame	5 million	Frame rate	Up to 1.2 fps
Optical configuration	Two cameras, one projector	Light source	Blue light
Color image output	Yes, RGB image registered to the point cloud	Data interface	Gigabit Ethernet
Power	24 V DC; power supply rated 75 W or more	Software	TridiVision
Dimensions L × W × H (mm)	166 × 90 × 54	Weight (kg)	0.87

¹ Regional Z repeatability, 2σ, measured by the method on page 22; the largest value found at the near, optimum and far working distance. ³ Optimum measuring volume: the depth over which the stated figures hold; the measuring depth is the full Z range over which points are returned (definitions, page 22).

VRH9 and VR series: technical data

VRH9 series

Model	VRH9-020B	VRH9-040B	VRH9-080B	VRH9-120B	VRH9-160B	VRH9-200B	VRH9-280B
Working distance, near (mm)	112	243	260	260	340	330	450
Working distance, optimum (mm)	113	250	270	275	360	360	500
Working distance, far (mm)	114.5	257	280	290	380	390	550
Field of view at near WD, W × H (mm)	24.1 × 12.4	47 × 24	79 × 40	120 × 61.5	151 × 77	190 × 98	251 × 129
Field of view at optimum WD, W × H (mm)	24.3 × 12.5	48.2 × 24.7	81.5 × 41.5	126 × 64.5	159 × 81.5	205 × 105	277 × 142
Field of view at far WD, W × H (mm)	24.6 × 12.6	49.5 × 25.5	84 × 43	132 × 67.5	167 × 85	220 × 112	302 × 155
Measuring depth, Z range (mm)	2.5	14	20	30	40	60	100
XY point spacing at optimum WD (μm)	5.8	11	19	30	38	49	66
Z repeatability, regional 2σ (μm) ¹	0.15	0.23	0.36	0.5	0.95	1.3	1.8

COMMON TO ALL MODELS OF THE SERIES			
3D points per frame	9.07 million (4200 × 2160)	Frame rate	1.1 fps at 4200 × 2160; 6.6 fps at 640 × 480 (guidance mode)
Optical configuration	One projector, one camera	Light source	Blue light
Color image output	No	Data interface	Gigabit Ethernet
Power	24 V DC; power supply rated 75 W or more	Software	TridiVision
Dimensions L × W × H (mm)	190 × 140 × 60	Weight (kg)	1.8

¹ Regional Z repeatability, 2σ, measured by the method on page 22; the largest value found at the near, optimum and far working distance.

VR series

Model	VR-400B	VR-600B	VR-1300B	VR-2300B
Working distance, near (mm)	350	500	900	1900
Working distance, optimum (mm)	500	700	1500	2600
Working distance, far (mm)	650	900	2200	3800
Field of view at near WD, W × H (mm)	215 × 160	300 × 225	530 × 395	1100 × 820
Field of view at optimum WD, W × H (mm)	300 × 220	415 × 305	865 × 645	1500 × 1115
Field of view at far WD, W × H (mm)	380 × 285	525 × 385	1250 × 940	2180 × 1625
Measuring depth, Z range (mm)	300	400	1300	1900
Optimum measuring volume, W × H × D (mm) ²	300 × 220 × 150	415 × 305 × 200	865 × 645 × 700	1500 × 1115 × 1200
XY point spacing at optimum WD (μm)	150	200	420	730
Z repeatability, regional 2σ (μm) ¹	2	14	60	100

COMMON TO ALL MODELS OF THE SERIES			
3D points per frame	3.2 million	Frame rate	1.1 fps at full resolution; 4 fps at 640 × 480 (guidance mode)
Optical configuration	One projector, one camera	Light source	Blue light
Color image output	No	Data interface	Gigabit Ethernet
Power	24 V DC; power supply rated 75 W or more	Software	TridiVision
Dimensions L × W × H (mm)	220 × 120 × 56	Weight (kg)	1.6

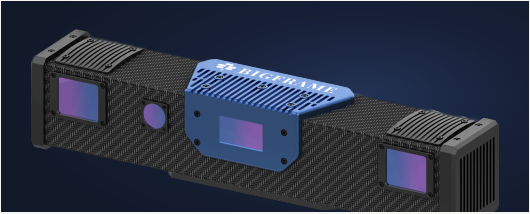
¹ Regional Z repeatability, 2σ, measured by the method on page 22; the largest value found at the near, optimum and far working distance. ² Optimum measuring volume: the depth over which the stated figures hold; the measuring depth is the full Z range over which points are returned (definitions, page 22).

PRS series: technical data

PRS series

NEW

Model	PRS-2500B
Working distance near / optimum / far (mm)	1200 / 2500 / 3500
Field of view at 1200 / 2500 / 3500 mm, W × H (mm)	1300 × 1050 / 2560 × 2060 / 3600 × 2907
Measuring depth, Z range (mm)	2300
XY point spacing at 1200 / 2500 / 3500 mm (μm)	680 / 1400 / 2000
Z repeatability (μm) ¹	300
Camera resolution	2448 × 2048 pixels, 5 MP, 2 cameras
Frame rate	1 fps at full resolution
Optical configuration	Two cameras, stereo baseline approx. 300 mm; laser pattern projector; RGB camera
Light source	Red laser, 638 nm
Laser class	Stated at launch
Color image output	RGB image registered to the point cloud
Data interface	Gigabit Ethernet
Power	24 V DC, up to 4 A
Operating temperature	0 to 45 °C
Dimensions L × W × H (mm)	890 × 97.5 × 95
Weight (kg)	3.9
Software	TridiVision



MEASURING FACE

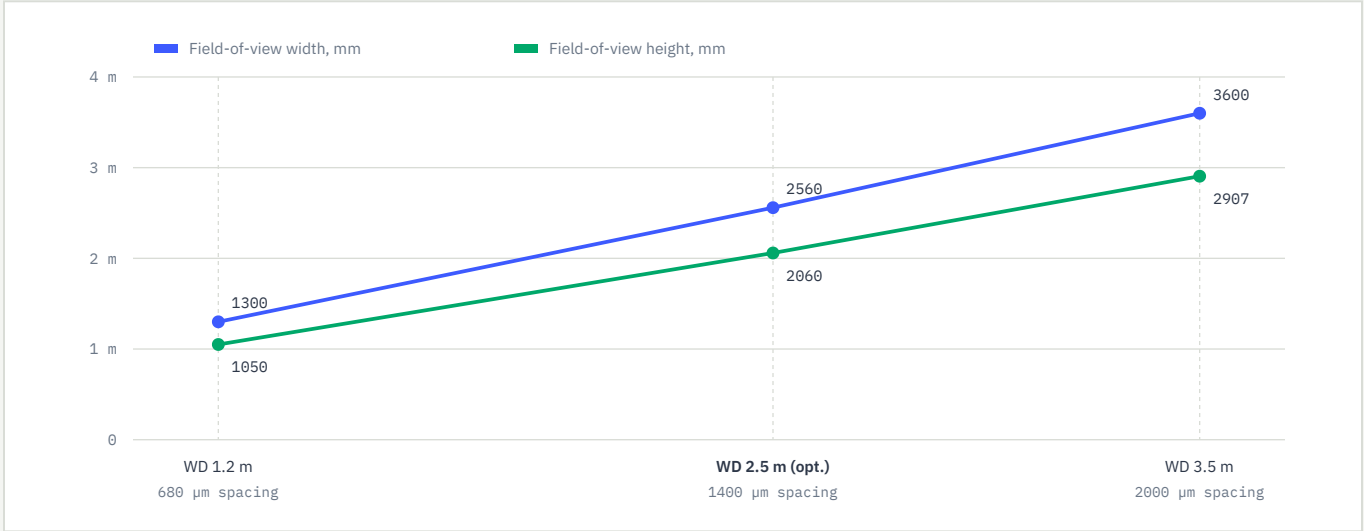
PRS, measuring face; housing 890 × 97.5 × 95 mm, camera windows at both ends, RGB camera and laser projector module between them.

By working distance

WD (mm)	Field of view (mm)	XY point spacing (μm)
1200 near	1300 × 1050	680
2500 opt.	2560 × 2060	1400
3500 far	3600 × 2907	2000

Preliminary specification; final data sheet at launch. ¹ Method to be stated at launch. Dimensions over the housing, connectors excluded.

Field of view and point spacing against working distance



Field-of-view width and height at the three specified working distances, with the XY point spacing under each. At 2.5 m the field of view covers a Euro pallet with margin; at 3.5 m it covers a container door.

Cables and calibration targets

Cables

Cable	Description	For series
VR-Power-I/O, 5 m	Power and I/O cable, 5 m	VRH9, VR, VRD
VR-Power-I/O, 10 m	Power and I/O cable, 10 m	VRH9, VR, VRD
VR-Power-I/O, 20 m	Power and I/O cable, 20 m	VRH9, VR, VRD
VR-Ethernet Cable, 5 m	Gigabit Ethernet cable, 5 m	VRH9, VR, VRD
VR-Ethernet Cable, 10 m	Gigabit Ethernet cable, 10 m	VRH9, VR, VRD
VR-Ethernet Cable, 20 m	Gigabit Ethernet cable, 20 m	VRH9, VR, VRD
TRS-CP12-HD-HD, 5 m	CoaXPress cable, HD-BNC to HD-BNC, 5 m	TRS
TRS-CP12-HD-HD, 10 m	CoaXPress cable, HD-BNC to HD-BNC, 10 m	TRS
TRS-CP12-HD-HD, 20 m	CoaXPress cable, HD-BNC to HD-BNC, 20 m	TRS

VRH9, VR and VRD share one power and I/O cable and one Gigabit Ethernet cable; pin assignment in the data sheet. Power supply 24 V DC, rated 75 W or more; the PRS draws up to 4 A. The TRS series outputs over CoaXPress and requires a CoaXPress frame grabber in the host computer; recommended grabber on request. Power and I/O cable for the TRS series and cables for the PRS: on request.



Power and I/O cable



Gigabit Ethernet cable

Calibration targets

Each camera model is calibrated on a target matched to its field of view: an array of square markers on a flat substrate. Marker pitch is the distance between marker centers, marker size the side of each square, pattern tolerance the position tolerance of the marker corners. Targets for the VRH9-020B, the TRS series and the PRS on request.

Model	Pattern	Pattern tolerance	Overall size	Array	Corner points	Marker pitch	Marker size	For camera models
CAL-CB-HC3D-40-6×6		±5 μm	40 × 40 mm	6 × 6	5 × 5	2.5 mm	1.8 mm	VRH9-040B, VRH9-080B
CAL-CB-HC3D-50-6×6		±5 μm	50 × 50 mm	6 × 6	5 × 5	6 mm	4 mm	VRH9-120B, VRH9-160B
CAL-CB-HC3D-90-6×6		±5 μm	90 × 90 mm	6 × 6	5 × 5	12 mm	9 mm	VRH9-200B, VRH9-280B
CAL-CB-HC3D-200-6×6		±10 μm	200 × 200 mm	6 × 6	5 × 5	25 mm	20 mm	VR-400B, VR-600B, VRD-600B
CAL-CB-HC3D-400-6×6		±10 μm	400 × 400 mm	6 × 6	5 × 5	60 mm	48 mm	VR-1300B, VRD-1200B
CAL-CB-HC3D-770-9×9		±10 μm	770 × 770 mm	9 × 9	8 × 8	80 mm	64 mm	VR-2300B

Targets and cables: order them with the camera.

Tell us the model and the cable lengths; you receive one quotation for camera, cables, target and software.

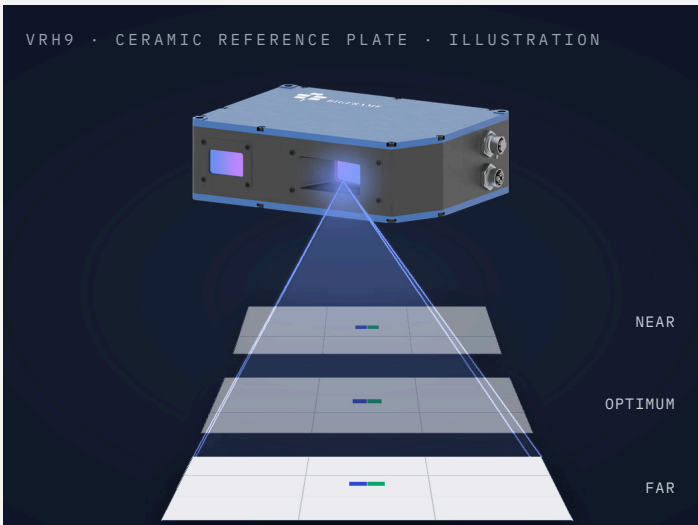
Definitions and method

Every figure in this catalog is stated with the condition under which it applies. Bigframe publishes repeatability, and the method with it.

Definitions

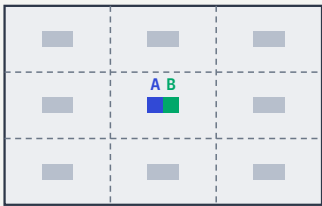
3D points per frame	The sensor grid: the maximum number of points one capture can return.
Working distance (WD)	Camera reference plane to the part. Every model is specified at the near, optimum and far working distance.
Field of view	Measured width × height at a stated working distance.
Measuring depth	Near to far working distance: the depth over which the camera returns points.
Optimum measuring volume	Field of view at the optimum working distance × the depth over which the stated figures hold.
XY point spacing	Distance between neighboring points at the optimum working distance: a sampling pitch, not the smallest resolvable feature.
Z repeatability, regional, 2σ	Repeatability of the height difference between two adjacent areas over 100 captures; the largest 2σ found is printed.
Frame rate	3D frames per second at the stated resolution.

Z repeatability: the method



The reference plate at the near, optimum and far working distance. Illustration.

A ceramic reference plate is imaged at the near, optimum and far working distance. The field of view is divided into nine regions; in each, two adjacent areas of under 1 % of the field are measured. The difference of their mean heights is measured 100 times; its standard deviation is the regional repeatability (1σ). Twice that value (2σ), the largest across nine regions and three distances, is the printed figure.



Nine regions; A and B adjacent in each
Reference plate, plan view

The figure is the repeatability of a difference of two area means: the standard deviation of $A - B$ over 100 captures (1σ), printed as 2σ . It is smaller than the repeatability of a single point, which is not published.

Why repeatability

Repeatability is one component of measurement uncertainty and can be stated for the camera alone, on a stated method. Whether a camera holds a tolerance is decided on the part, by a gauge study; the figures here select the model to start with.

Where the figures are

Series pages 5–14: models at a glance and fields of view. Page 17: all eighteen models on one table. Pages 18–20: full specification per series; page 21: cables and calibration targets.



The company

Bigframe Technology Co., Ltd. designs and manufactures structured-light 3D cameras and the TridiVision software that runs them. Founded in 2021 and based in Kunshan, Suzhou, the company continues the 3D vision work begun in 2015 by Imalligent Technology (Shanghai), now its wholly owned subsidiary. Its eighteen camera models in five series measure, inspect and guide on production lines in consumer electronics, semiconductor packaging, batteries, automotive and logistics.

Every camera except the preliminary PRS is characterized by the same published method — regional Z repeatability, 2σ , on a ceramic reference plate at three working distances.

Contact

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WHERE THE CAMERAS WORK

Consumer electronics

connectors, mid-frames, adhesive beads, printed circuit assemblies

Semiconductor packaging

solder balls, wafer bumps, lead frames

Batteries and electric vehicles

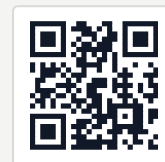
cell contact system welds, module assembly

Automotive

body panels, castings, machined parts

Logistics and general automation

bin picking, palletizing, tray inspection, rail infrastructure



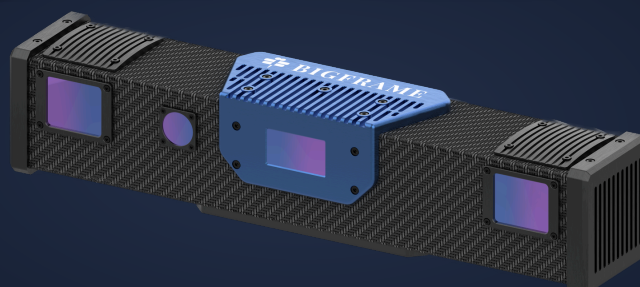
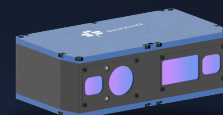
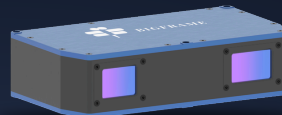
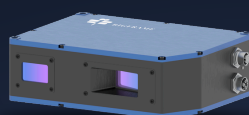
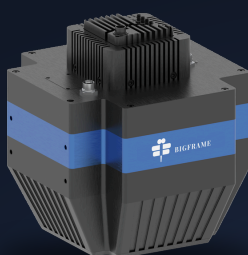
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Notes

Specifications are subject to change without notice. Dimensions are over the housing, connectors excluded; weights are without cables. Product images are renderings. Z repeatability is the regional 2σ value measured by the method on page 22. Application images show height maps, point clouds and color images from Bigframe application cases; the depalletizing scene on page 4 is a rendering. TridiVision is a trademark of Bigframe Technology Co., Ltd. NVIDIA and CUDA are trademarks of NVIDIA Corporation. CoaXPress is a trademark of the Japan Industrial Imaging Association. Catalog EN 2026-10 · REV. C · © 2026 Bigframe Technology Co., Ltd.



L-R: TRS · VRH9 · VR · VRD · PRS (FRONT)



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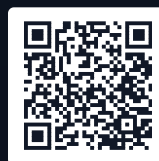
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